

OXFORD

INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL A-LEVEL CHEMISTRY

(9620) CH05

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH05 January 2024

QUESTION 01

01.1 Many answered in terms of keeping the temperature constant, but the question clearly says BEFORE each experiment. This prevents the situation where reagents are stored in different places having different temperatures.

01.2 Most completed the calculation correctly but some lost a mark for incorrect rounding.

01.3 Most got this correct. Some students referred to the mixture boiling, the iodine decomposing, safety issues and not being able to reach that temperature with a water bath.

01.4 Many misread the instructions and plotted all their points. This mistake was not penalised. This meant that a smaller than expected scale was needed for the graph. Again, this was not penalised. Some unusual scales were chosen for the y-axis. This made plotting more challenging. A few students plotted one or more points incorrectly, even on a sensible scale. In particular, this applied to point 3 where there was confusion as to where 2.99×10^{-3} should be. It is recommended that a pencil is used for drawing graphs, so that an error can be erased.

01.5 Some took advantage of the alternative data provided even though they had the correct gradient from their graph. This was not a problem. Some clearly knew what was needed, some failed to convert to kJ mol^{-1} but some had little idea of what to do. Once again, the way that some students form their number characters on the paper highlighted some problems. Some characters looked sufficiently like “93.9” in a calculation that could give no other answers, so credit could be given but it would be difficult to do so in other circumstances.

01.6 Many gave the order with respect to $[\text{I}^-]$ as 2. They lost that mark but that error was carried forward so they could gain M4 and, possibly, M5.

QUESTION 02

02.1 Most got “silver mirror formed”. For M2 and M3, a few got them the wrong way round. Many ignored the instruction to not use butanal or butanone. Some made the error of using $-\text{COH}$ rather than $-\text{CHO}$ to show an aldehyde.

02.2 In this experiment, being able to control the reaction is important but maintaining a constant temperature is not.

02.3 Most got the “red” precipitate. Some missed out the precipitate (or solid). A few confused Fehling’s and Tollens’ and some appeared to have chosen any one of the identification tests in the specification.

02.4 Most got 4 and 6 respectively. 2 and 3 was a common error, perhaps the students incorrectly saw additional symmetry.

QUESTION 03

03.1 This was generally done well. A few students added either one of the components to the diagram but not the other. Some missed the question even though they had clearly seen the diagram (they had ringed parts of the labels) but didn’t add anything to the diagram.

03.2 Most got this correct but a significant number chose 101 kPa as the standard pressure.

03.3 Unless Cu^{2+} was part of the answer for M1, M2 could not be awarded. If both Cu^{2+} and Cu were in M1, without clarity, M2 was accessible.

03.4 Lots of students discussed hydrolysis. Some had flawed calculations but if they identified that the problem was $[\text{Al}^{3+}]$ not being 1.0 mol dm^{-3} they were given the mark.

03.5 “ BaSO_4 is a solid” was not sufficient since some solids are soluble. “Precipitate” was sufficient because it implies insolubility. “Barium is reactive” was not sufficient – “very” or “reacts with water” was needed.

QUESTION 04

This was a fairly easy question. There was a spread across the distractors, with C being the most popular.

QUESTION 05

Just under half of the students got this correct. B was a popular distractor.

QUESTION 06

This was a fairly easy question. There was a reasonably even spread across the distractors.

QUESTION 07

This was a fairly easy question. A and B were the popular distractors.

QUESTION 08

This was a fairly easy question. A was the most popular wrong answer.

QUESTION 09

This was a very easy question.

QUESTION 10

This was a very easy question.

QUESTION 11

This was a very easy question.

QUESTION 12

This was a fairly easy question. C was a popular incorrect choice.

QUESTION 13

This was a fairly easy question although B attracted some students.

QUESTION 14

This was a fairly easy question. C was a popular incorrect choice.

QUESTION 15

Just over half of the students got this question correct. C and D were almost equally attractive distractors.

QUESTION 16

This was a fairly easy question. D was attractive to some students.

QUESTION 17

This was a fairly easy question. C attracted some students.

QUESTION 18

Just over half of the students got this question correct. Most incorrect students chose B.

QUESTION 19

Just over half of the students got this question correct. D was a very popular wrong answer.

QUESTION 20

This was a fairly easy question. C attracted a few students.

QUESTION 21

This was a fairly easy question. D was the least attractive distractor.

QUESTION 22

This was a fairly easy question. The distractors were more or less equally attractive.

QUESTION 23

Around 60% of students got this question correct. A was a popular incorrect answer.

QUESTION 24

This was a very easy question.

QUESTION 25

This was a fairly easy question. B attracted a few incorrect students.

QUESTION 26

This was a fairly easy question. There was an even spread across the incorrect options.

QUESTION 27

This was a fairly easy question. B attracted a significant number of incorrect responses.

QUESTION 28

This was a fairly easy question. A few chose D.

QUESTION 29

This was a fairly easy question. C was the most popular incorrect answer.

QUESTION 30

This was a fairly easy question. D was the most popular distractor.

QUESTION 31

This was a fairly easy question. B attracted a few students.

QUESTION 32

This was a fairly easy question. A and C were the popular distractors.

QUESTION 33

Approximately equal numbers of students went for distractor D as for the correct answer B.

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